

Bhavana Macha

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EDUCATION

Master's in Data Analytics Engineering

George Mason University • Fairfax, Virginia • 2023 • 3.5

Bachelor's in Electronics & Communications Engineering

Jawaharlal Nehru Technological University • Hyderabad, India • 2020 • 3.5

SKILLS

- **Programming languages:** Python, HTML, CSS, JavaScript, SQL, R.
- **Databases:** Mongo DB, SQLite, MySQL, PostgreSQL
- **Tools:** Tableau, RStudio, Anaconda, VS Code, Oracle, Power BI, MS Excel, Big ML, RapidMiner, Knime, Orange, Hadoop.

EXPERIENCE

Data Analyst Intern

George Mason University in partnership with MITRE corporation

January 2023 - May 2023, Fairfax, VA

- Successfully integrated OpenMetadata with Elastic Search to establish a comprehensive data catalog system, enabling efficient search and discovery of data assets. Used an S3 bucket as a centralized repository for data assets.
- Implemented data ingestion pipelines to automatically extract metadata from various data sources and populate the data catalog.
- Prepared comprehensive documentation, including architectural diagrams, configuration details, and integration process documentation.

Data Analyst Intern

SForce IT Solutions

June 2022 - August 2022, Texas

- Acquired data from primary or secondary data sources and maintained databases/data for the department of International and Domestic Marketing by using SQL and MS Excel.
- Performed cleaning and pre-processing activities on the data sets using Python and pandas.
- Used Tableau to create dashboards to monitor key metrics and provide executive-level summaries of data.
- Launched a project to improve data reporting (40% efficiency gain) by automating the process of data extraction and creating a user-friendly interface using R.

PROJECTS

Data mining project on a credit card data set

George Mason University • August 2022 - December 2022

- Completed Data mining project on a credit card data set using machine learning methods such as Decision trees, KNN, and Random Forests, which increased target positive rate.

Electric vehicles dataset

George Mason University • September 2021 - December 2022

- Created dashboards to help people decide what EVs to go for based on a number of metrics including performance, pricing, and range per charge.
- Publication: <https://public.tableau.com/app/profile/akshay.munjal/viz/Electric-Vehicle-Blog/Story1?publish=yes>

Detection of flight delay using ML

George Mason University • August 2022 - November 2022

- Removed null values, normalized outliers, and split date columns into day, month, and hour to clean the data. Decreased model's Mean absolute percentage error to 0.4.
- Implemented Random Forest algorithm with scikit-learn to detect delays in airline flights. Tuned the model according to the validation set to determine how many minutes the flight is going to be delayed.
- Performed exploratory data analysis to identify key attributes affecting the delay of flights.

Detection of cancer using ML

JNTUH • December 2019 - April 2020

- Trained and tested the ML model with millions of records of patients with cancer. The data was collected from a real hospital.
- Used a different predictive algorithm on the data and compared the accuracy. Random forest model gave the best accuracy and hence best predictions.
- Written a research paper on the architecture, results and how the model can be further improved.

CERTIFICATIONS

Data mining

Linkedin Learning • 2022

Advanced Python: working with databases

Linkedin Learning • 2022